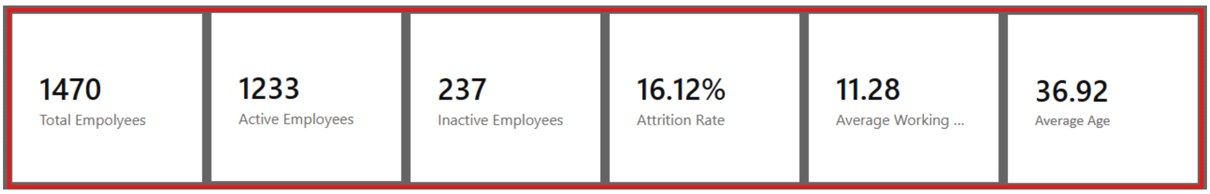
# Power BI Dashboard & Visualizations Guide


Dashboard

## 1. Executive Dashboard Overview

The main Power BI dashboard acts as the central hub for HR stakeholders to monitor the health of the organization's workforce. It is designed with a dark, modern aesthetic to make high-contrast data stand out clearly.

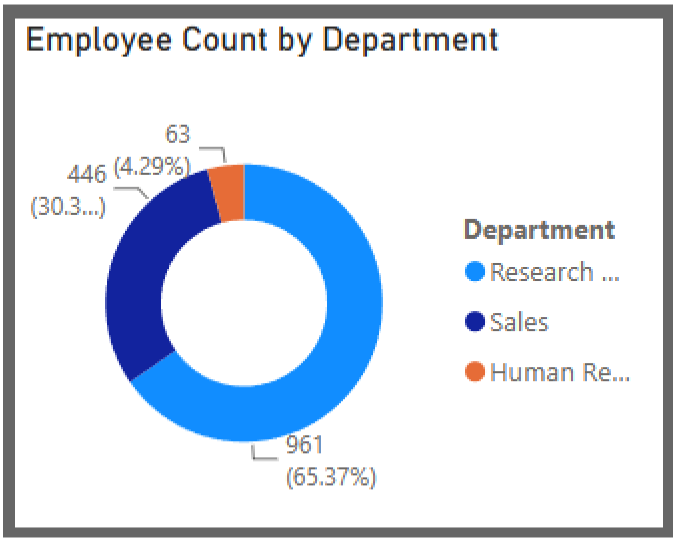
### Top-Level KPI Cards




- **What it does:** Displays the most critical snapshot metrics at the very top of the page.
- **Metrics Tracked:** Total Employees (1470), Active Employees (1233), Inactive Employees (237), Overall Attrition Rate (16.12%), Average Working Years (11.28), and Average Age (36.92).
- **Purpose:** Provides immediate, high-level context before diving into the granular charts below.

# 2. Core Demographic & Role Visualizations

### Employee Count by Department & Gender




- **Visual Type:** Donut Charts.
- **What it does:** Breaks down the raw headcount of the company by overarching departments (e.g., R&D, Sales) and gender identity.
- **Purpose:** Establishes the baseline distribution of the workforce to help HR see if attrition in a specific group is disproportionate to its overall size.

### Attrition by Job Role and Satisfaction

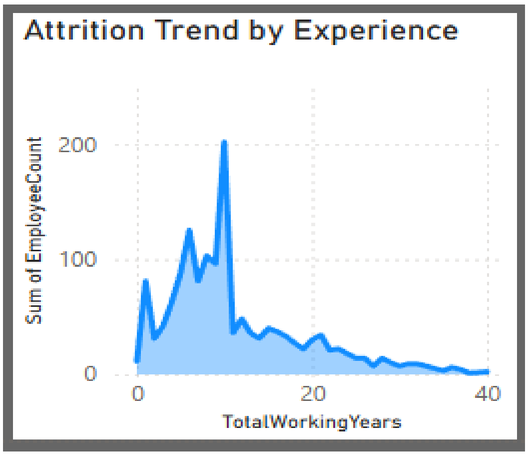
Attrition by Job Role and Satisfaction

| JobRole | 1 | 2 | 3 | 4 | Total |
|---|---|---|---|---|---|
| Laboratory Technician | 20 | 8 | 21 | 13 | **62** |
| Sales Executive | 16 | 9 | 18 | 14 | **57** |
| Research Scientist | 13 | 10 | 15 | 9 | **47** |
| Sales Representative | 7 | 10 | 9 | 7 | **33** |
| Human Resources | 5 | 2 | 3 | 2 | **12** |
| Manufacturing Director | 2 | 2 | 4 | 2 | **10** |
| Healthcare Representative | 2 | 2 | 1 | 4 | **9** |
| Manager | 1 | 2 | 1 | 1 | **5** |
| Research Director | | 1 | 1 | | **2** |
| **Total** | **66** | **46** | **73** | **52** | **237** |

- **Visual Type:** Matrix Table.
- **Fields Used:** `JobRole` (Rows), `JobSatisfaction` (Columns), Count of `Attrition` (Values).
- **What it does:** Cross-references the specific job titles of departing employees with the job satisfaction score they reported before leaving.
- **Business Value:** Identifies hidden flight risks. For example, it reveals that Laboratory Technicians and Sales Executives frequently leave despite reporting high (Level 3 or 4) job satisfaction, indicating they are likely leaving for external reasons like better pay.

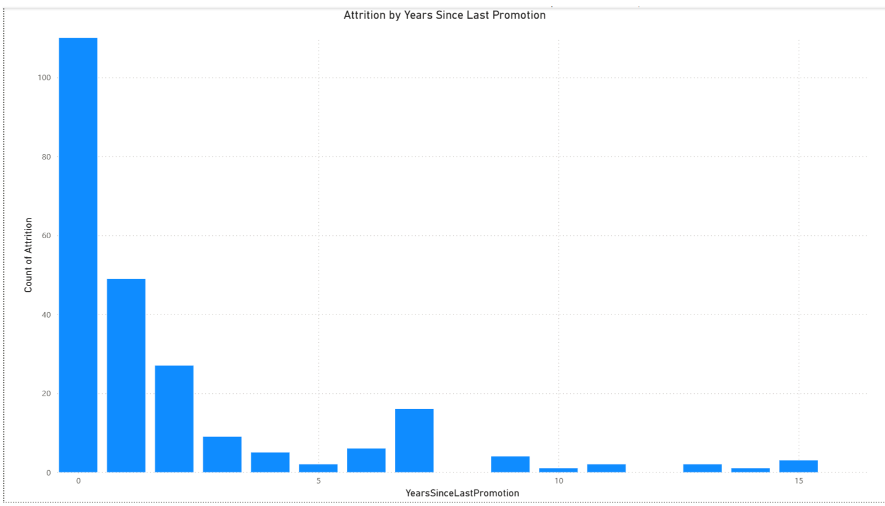
# 3. Behavioral & Timeline Visualizations (Deep Dives)

### Attrition Trend by Experience (Total Working Years)



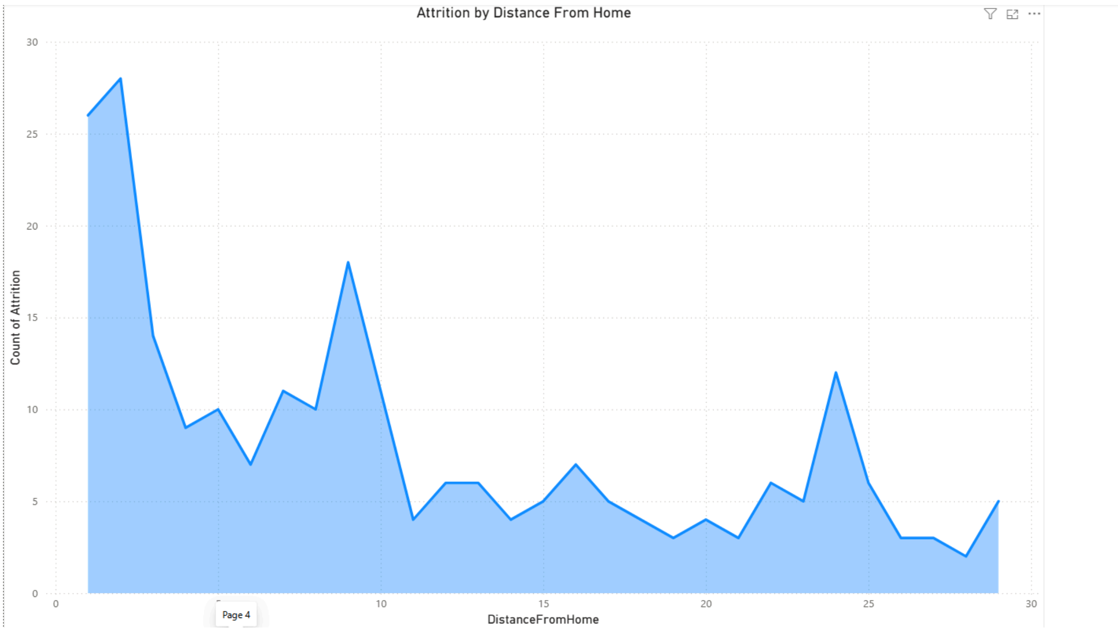

- **Visual Type:** Area Chart.
- **What it does:** Maps the volume of employee departures along a continuous timeline of their total career experience.
- **Business Value:** Highlights a massive "danger zone" in the first 1 to 5 years of an employee's tenure. It shows management exactly when employees are most vulnerable to quitting.

### Attrition by Years Since Last Promotion



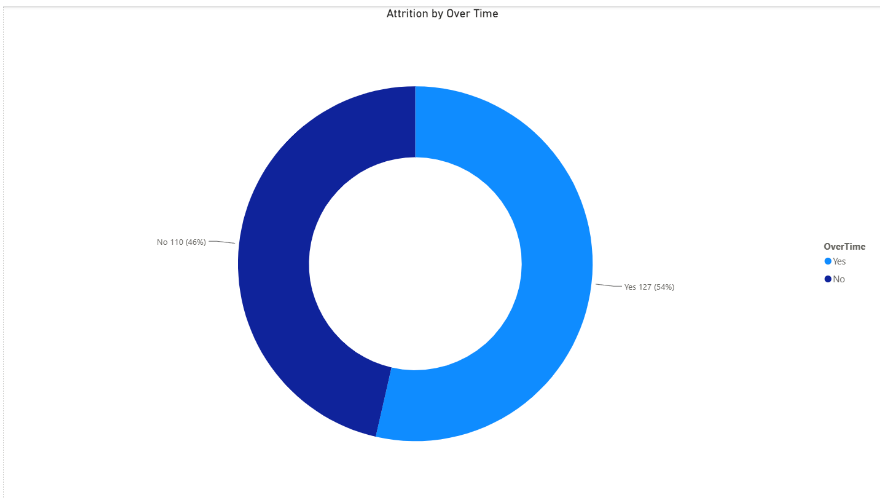

- **Visual Type:** Clustered Column Chart.
- **Fields Used:** `YearsSinceLastPromotion` (X-axis), Count of `Attrition` (Y-axis).
- **What it does:** Groups departing employees by how long they remained in a role without a step up.
- **Business Value:** Identifies career stagnation. A massive spike at the "0" mark shows employees leaving immediately after a promotion (likely leveraging their new title elsewhere), while a secondary spike around year 7 shows the breaking point for stagnant employees.

### Commute Fatigue: Attrition by Distance From Home



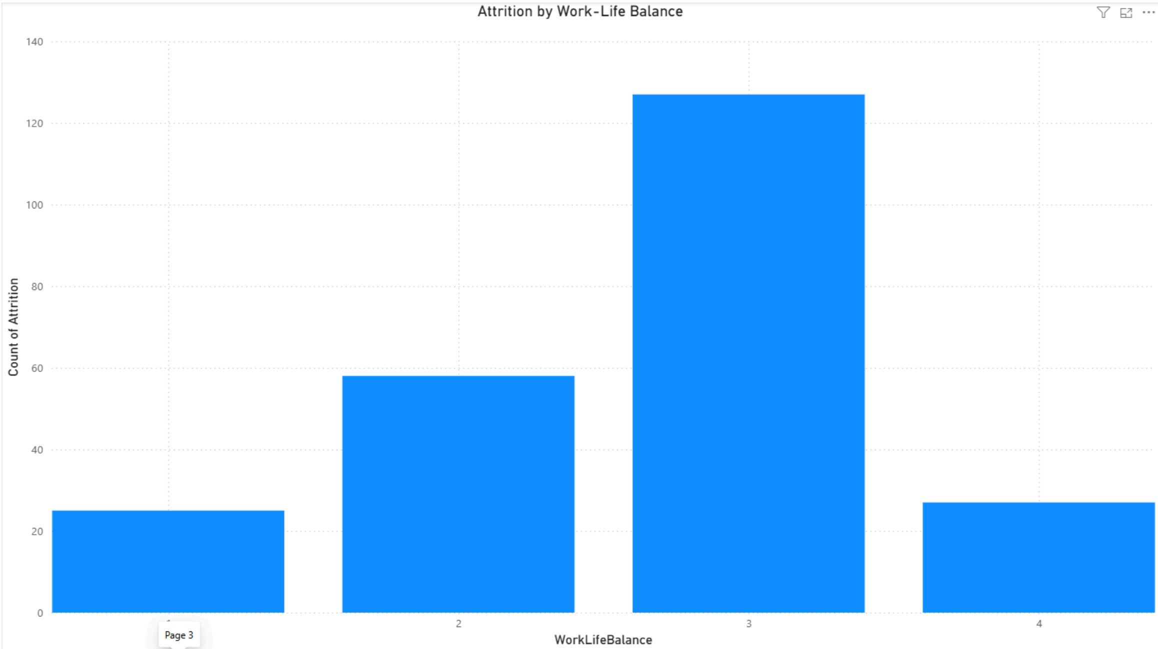

- **Visual Type:** Area Chart.
- **Fields Used:** `DistanceFromHome` (X-axis), Count of `Attrition` (Y-axis).
- **What it does:** Visualizes the daily commute distance of employees who quit.
- **Business Value:** Reveals spikes in turnover at the 9-10 mile and 24-25 mile marks, providing HR with data to justify remote-work or hybrid-schedule interventions for employees with long commutes.

### The Burnout Factor: Attrition by OverTime



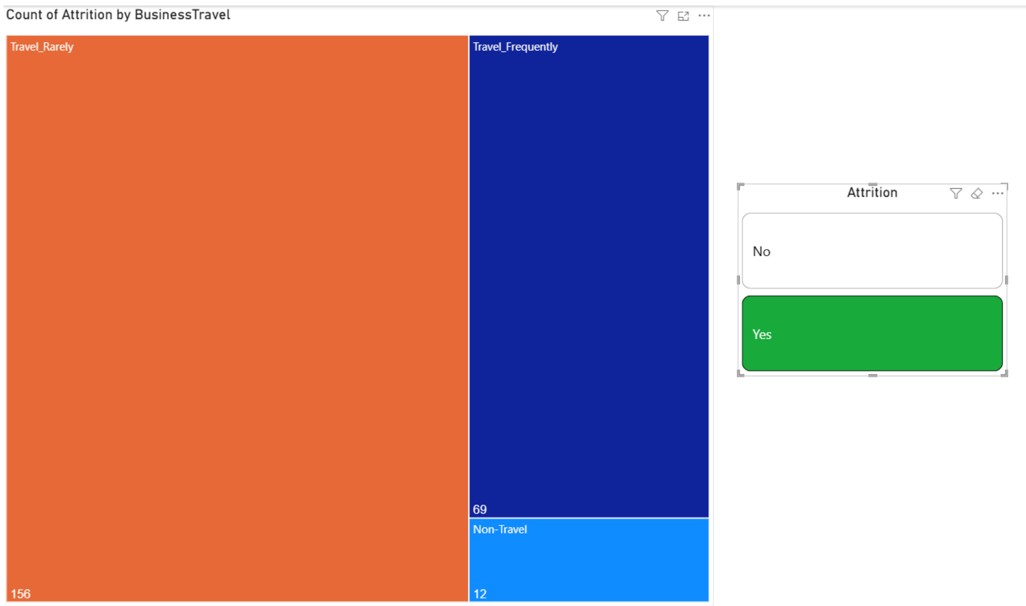

- **Visual Type:** Donut Chart.
- **Fields Used:** `OverTime` (Legend), Count of `Attrition` (Values).
- **What it does:** Compares the proportion of departing employees who worked overtime versus those who did not.
- **Business Value:** Highlights that 54% of turnover comes from overtime workers, proving that excess workload is a primary driver of attrition.

### The Work-Life Balance Paradox




- **Visual Type:** Clustered Column Chart.
- **Fields Used:** `WorkLifeBalance` (X-axis), Count of `Attrition` (Y-axis).
- **What it does:** Shows the volume of departures across the 1 (Bad) to 4 (Best) Work-Life Balance rating scale.
- **Business Value:** Proves that a "Better" (3) work-life balance does not guarantee retention. The highest volume of departures came from this group, showing that scheduling flexibility cannot compensate for other underlying job issues.

### Travel Strain: Attrition by Business Travel



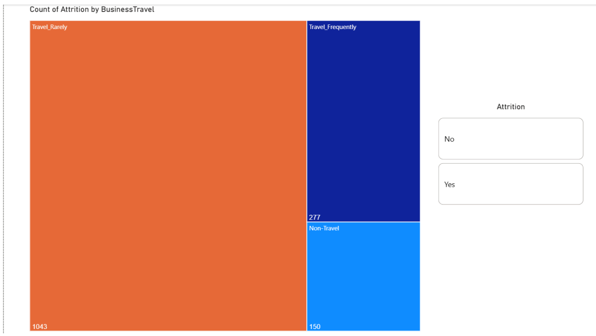

- **Visual Type:** Treemap.
- **Fields Used:** `BusinessTravel` (Category), Count of `Attrition` (Values).

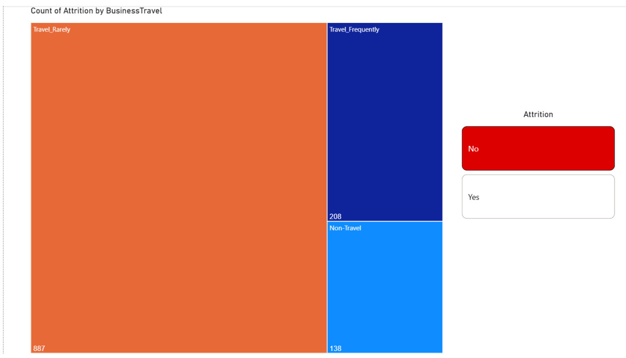
- **What it does:** Uses sized blocks to represent the volume of departures based on travel frequency (Non-Travel, Travel Rarely, Travel Frequently).
- **Business Value:** Allows stakeholders to quickly see the volume impact of travel. While "Travel Rarely" has the largest physical block (highest raw volume), "Travel Frequently" takes up a concerning amount of space relative to how few employees actually travel frequently.

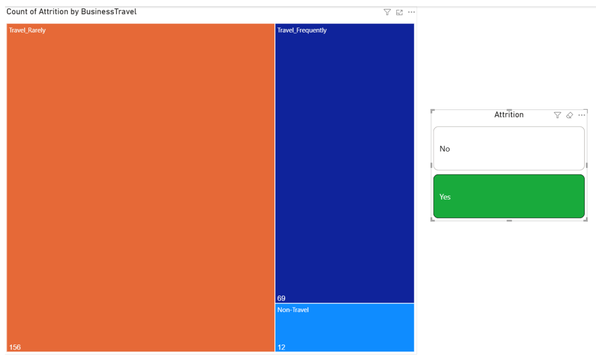
# 4. UI / UX Interactive Features

### Dynamic Attrition Slicer (State-Based Buttons)

- **Visual Type:** Slicer / Buttons with Conditional Formatting.
- **How it works:** The dashboard utilizes an interactive slicer to filter the page by Attrition ("Yes" or "No"). Using Power BI's advanced button states and rule-based conditional formatting, the buttons remain a clean **White** by default. When a user selects "Yes", the button dynamically turns **Green**, and when selecting "No", the button turns **Red**.




- **Purpose:** Enhances the user experience (UX) by providing immediate, intuitive visual feedback to the stakeholder interacting with the dashboard.